MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (FOUR PLIES, HEAVY SPIRAL WRAPPED WIRE, HIGH PRESSURE SIZES -8 THROUGH -32)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by four plies of heavy spiral wrapped wire reinforcement in sizes -8 through -32 for high pressure applications.

1.2 Classification. End fittings for hoses covered by this specification shall be in sizes -8 through -32 as applicable and shall be of the following types, as shown on figures 1 through 9:

Type AO	- SAE 37-degree tube swivel to hose, straight (see figure 1).
Type BO	- Gasket seal union, male half to hose, straight (see figure 2).
Type BLO	- Gasket seal union, male half to hose, 90-degree elbow (see figure 3).
Type CO	- O-ring seal union, male half to hose, straight (see figure 4).
Type CLO	- O-ring seal union, male half to hose, 90-degree elbow (see figure 5).
Type DO	- 90-degree elbow, hose to hose (see figure 6).
Type GO	- 180-degree return, hose to hose (see figure 7).
Type SAE J518c(A)	- O-ring flange, straight, with solid code 62 flange (see figure 8).
Type SAE J518c(B)	- O-ring flange, 90-degree bend with solid code 62 flange (see figures 9 and 10) (90-degree bend shown, also available with 22-1/2 degree, 45-degree, 60-degree and 67-1/2 degree tube bends).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 220 - Standard Specification for Pearlitic Malleable Castings. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 6381 - Steel Tubing, Mechanical 0.95Cr - 0.20Mo (0.38 - 0.43C) (SAE 4140). (DoD adopted)

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. The rubber compounds used shall be resistant to petrol base oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 9.

3.3 Reinforcement. The reinforcement shall consist of four plies of heavy spiral wrapped wire applied in such a manner as to meet all the requirements of this specification.

3.4 Dimension and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash numbers, performance characteristics, and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)						Min bend radius (inches)	Approx wt/ft (lbs/ft)
	Working (max)	Burst (min)	Id tube		Od cover		Od reinf			
			min	max	min	max	min	max		
-8	7,500	30,000	0.50	0.531	0.952	0.985	0.862	0.877	8	0.75
-12	6,250	25,000	.75	.781	1.242	1.275	1.128	1.152	11	1.05
-16	5,000	20,000	1.00	1.031	1.492	1.524	1.354	1.386	14	1.31
-20	4,000	16,000	1.25	1.281	1.775	1.807	1.611	1.659	16.5	2.05
-24	3,000	12,000	1.50	1.531	2.090	2.122	1.898	1.946	20	2.35
-32	3,000	12,000	2.00	2.031	2.661	2.693	2.460	2.500	28	3.13

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3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-16	4
-32	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

TABLE III. Numbers of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-16	8
-32	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-16	2	2	-16	4	4
-32	3	2	-32	5	5

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-16	2	1	1	-16	4	2	2
-32	3	1	1	-32	5	3	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers								
	Size (dash no.) -16				Size (dash no.) -32				
Test sequence	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X			
Impulse test (see 4.6.5)1/				X				X	X
Burst test (see 4.6.6)1/	X				X				
Cold temperature flexing (see 4.6.8)1/			X				X		

1/ See MIL-H-24135.

4.3 The test fluid shall be petroleum base oil in accordance with MIL-L-17331 or MIL-L-2104 for the aging, hydraulic fluid circulation, and impulse tests.

4.4 The peak pressure for the dynamic impulse test shall be 150 percent of the maximum working pressure (see table I).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

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6. NOTES

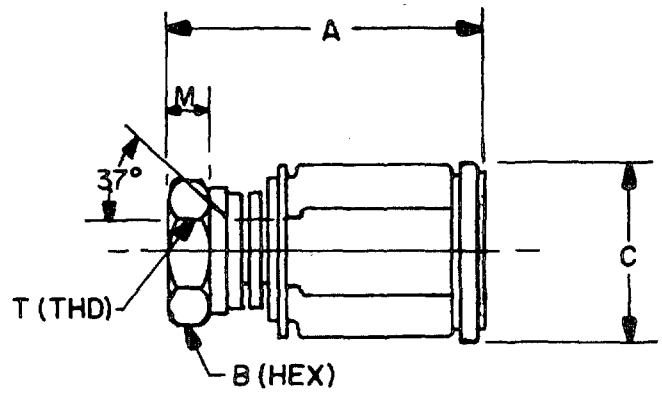
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems where pressures do not exceed the rated working pressure of the hose and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4720-N530-9)

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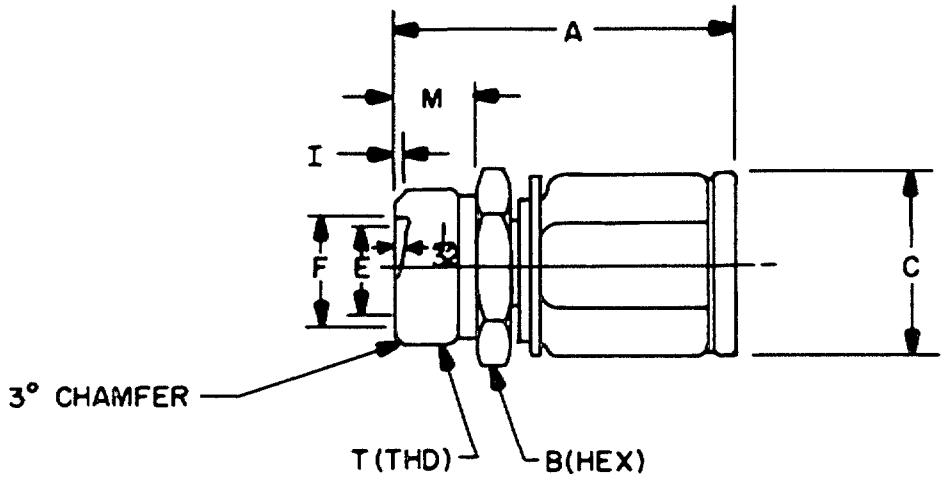
Tube size (nom)	Rubber hose (dash no.)	T thread (UN) CL-2B	A (max)	B (min)	C (max)	M (ref)
1/2	-8	3/4-16	3.857	1.12	1.44	0.44
3/4	-12	1-1/16-12	4.489	1.25	1.73	.57
1	-16	1-5/16-12	5.39	1.50	2.16	.60
1-1/4	-20	1-5/8-12	5.10	2.00	2.88	.64
1-1/2	-24	1-7/8-12	5.63	2.25	3.16	.74
2	-32	2-1/2-12	6.45	2.88	3.75	.93

NOTES:

1. Dimensions are in inches
2. Fitting assembly materials for the swivel nut and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Sockets for sizes -8, -12 and -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 1. Type AO, 37-degree tube swivel to hose, straight.

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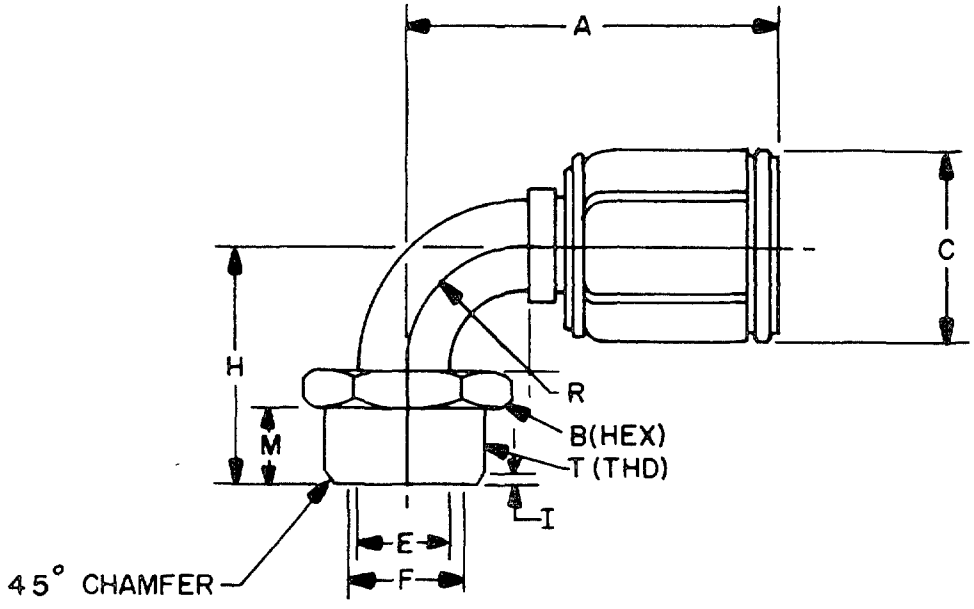
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (see note 2)	C (max)	E ± 0.005	F ± 0.005	I (ref)	M (ref)
1	-16	2-1/8-12 UN	5.42	2.250	2.16	1.093	1.375	5/64	1.12
1	-20	2-1/8-12 UN	5.36	2.250	2.88	1.093	1.375	5/64	1.12
1-1/4	-20	2-5/8-12 UN	5.61	2.750	2.88	1.344	1.750	3/32	1.25
1-1/4	-24	2-5/8-12 UN	5.92	2.750	3.16	1.344	1.750	3/32	1.25
1-1/2	-24	2-7/8-12 UN	6.11	3.000	3.16	1.656	2.094	7/64	1.31
2	-32	3-5/8-12 UN	6.61	3.750	3.75	1.937	2.375	1/8	1.38

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Socket for size -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 2. Type BO, gasket seal union, male half to hose, straight, tube sizes 1 to 2 inches.

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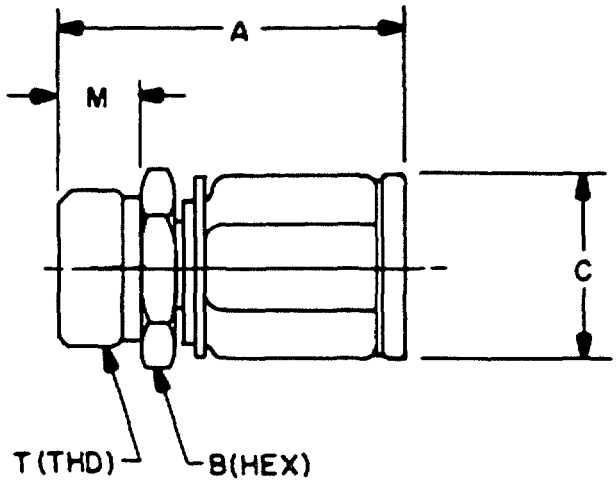
Nps	Rubber hose (dash no.)	T thread CL-3	A (max)	B (see note 2)	C (max)	E ± 0.005	F ± 0.005	H (max)	I (ref)	M (ref)	R (rad) (min)
1	-16	2-1/8-12 UN	6.08	2.250	2.16	1.093	1.375	3.18	5/64	1.12	1.50
1-1/4	-20	2-5/8-12 UN	6.44	2.750	2.88	1.344	1.750	3.83	3/32	1.25	1.75
1-1/2	-24	2-7/8-12 UN	7.25	3.000	3.16	1.656	2.094	4.52	7/64	1.31	2.00
2	-32	2-5/8-12 UN	8.95	3.750	3.75	1.937	2.375	5.57	1/8	1.38	3.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Socket for size -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 3. Type BLO, gasket seal union, male half to hose, 90-degree elbow, tube sizes 1 to 2 inches.

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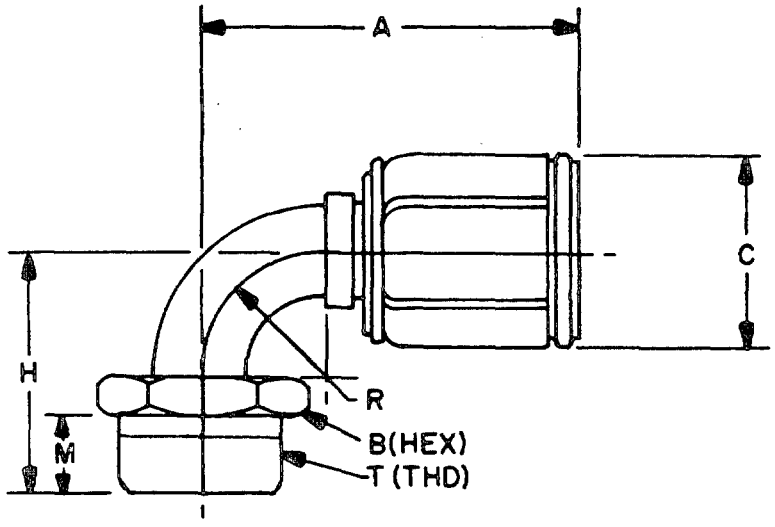
Nps	Hose end size (dash no.)	T thread CL-3A	A (max)	B (see note 2)	C (max)	M (ref)
3/8	-8	1-3/8-12	3.63	1.43	1.44	0.62
1/2	-12	1-3/4-12	4.41	1.81	1.73	.62
1	-16	2-5/16-12	5.17	2.38	2.16	.75
1	-20	2-5/16-12	5.10	2.38	2.88	.75
1-1/4	-20	2-3/4-12 UN	5.10	2.88	2.88	.75
1-1/4	-24	2-3/4-12 UN	5.42	2.88	3.16	.75
1-1/2	-24	3-1/16-12 NS	5.68	3.12	3.16	.88
2	-32	3-3/4-12 UN	6.01	3.88	3.75	.88

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Socket for sizes -8, -12 and -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 4. Type CO, O-ring seal union, male half to hose, straight, tube sizes 3/8 to 2 inches.

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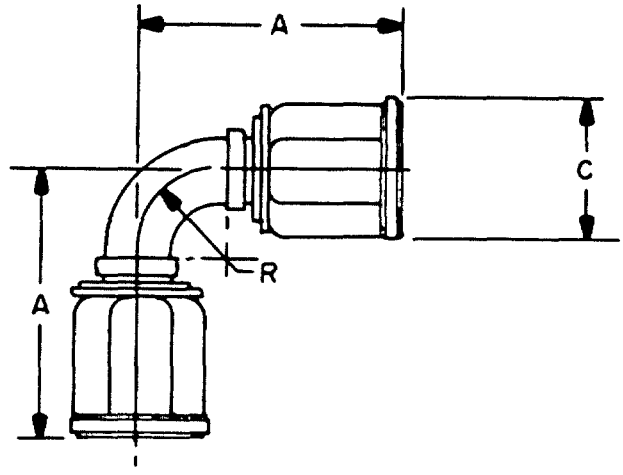
Tube size (nps)	Hose end size (dash no.)	T thread CL-3A	A (max)	B (see note 2) (min)	C (max)	H (max)	M (ref)	R (rad)
3/8	-8	1-3/8-12	4.45	1.44	1.44	1.95	0.62	0.75
1/2	-16	1-3/4-12	5.66	1.81	1.73	2.39	.62	1.25
1	-16	2-5/16-12 NS	6.08	2.38	2.16	3.01	.75	1.50
1-1/4	-20	2-3/4-12 UN	6.44	2.88	2.88	3.32	.75	1.75
1-1/2	-24	3-1/16-12 NS	7.25	3.12	3.16	3.95	.88	2.00
2	-32	3-3/4-12 UN	8.95	3.88	3.75	4.94	.88	2.50

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the male half union adapter and hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Socket for sizes -8, -12 and -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 5. Type CLO, O-ring seal union, male half to hose, 90-degree elbow, tube sizes 3/8 to 2 inches.

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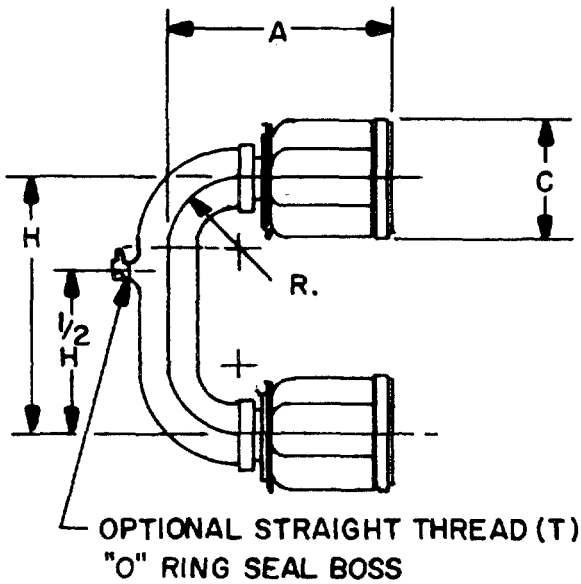
Rubber hose (dash no.)	A	C	R
-8	4.45	1.44	0.75
-12	5.66	1.73	1.25
-16	6.26	2.16	1.50
-20	6.44	2.83	1.75
-24	7.25	3.16	2.00
-32	8.95	3.75	3.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions. Socket for sizes -8, -12 and -16 shall be one-piece design.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 6. Type DO, 90-degree elbow hose to hose, hose dash sizes -8 through -32.

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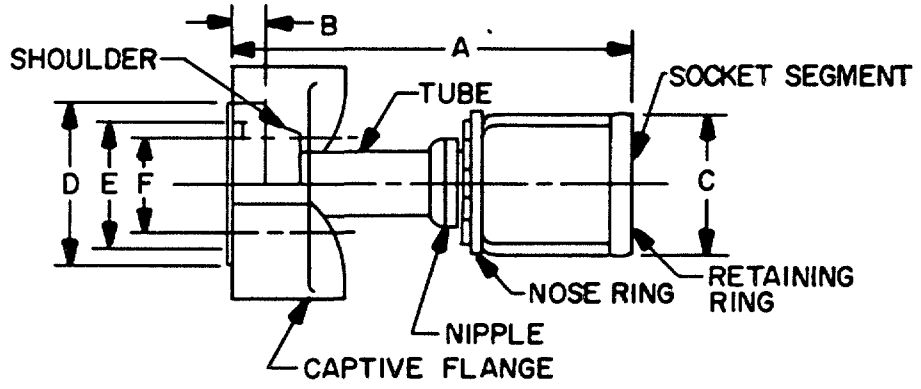
Rubber hose (dash no.)	A (max)	C (max)	H (max)	R (rad) (min)	Optional T thread UNF-2B
-24	7.25	3.16	6.50	2.00	5/16-24

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -24 hose may be the one-piece or multiple segment design, not exceeding the envelope dimensions.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose size -24 shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.

FIGURE 7. Type GO, 180-degree return, hose to hose, hose dash size -24.

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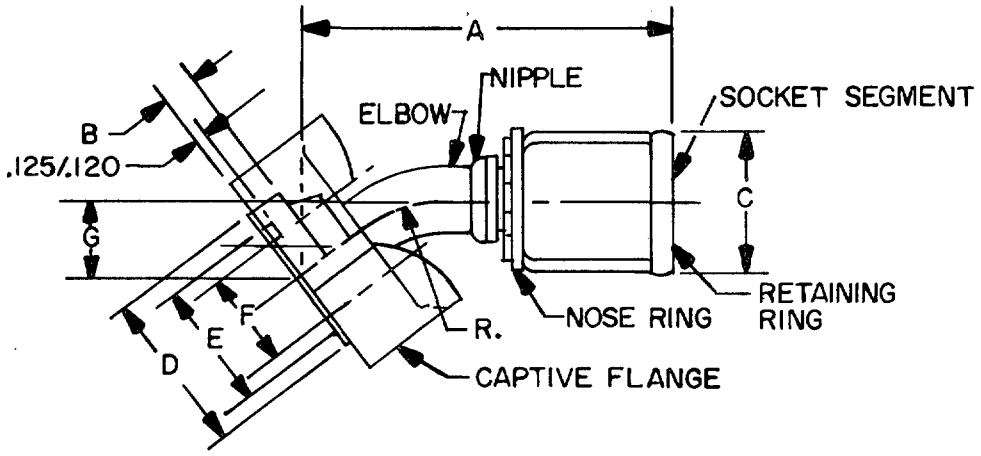
Size		Shoulder size	A	B	C	D	E	F
Od tube	Hose							
1-1/2	20	-20	7.34	0.405	2.88	2.125	1.752	1.422
1-1/2	24	-24	7.65	.495	3.16	2.500	2.120	1.890
2	32	-32	8.94	.495	3.75	3.125	2.495	2.165

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.
7. Captive flange material shall be malleable iron in accordance with ASTM A 220, grade 50005.

FIGURE 8. Type SAE J518c, flange fitting, straight, with code 62 solid captive flange.

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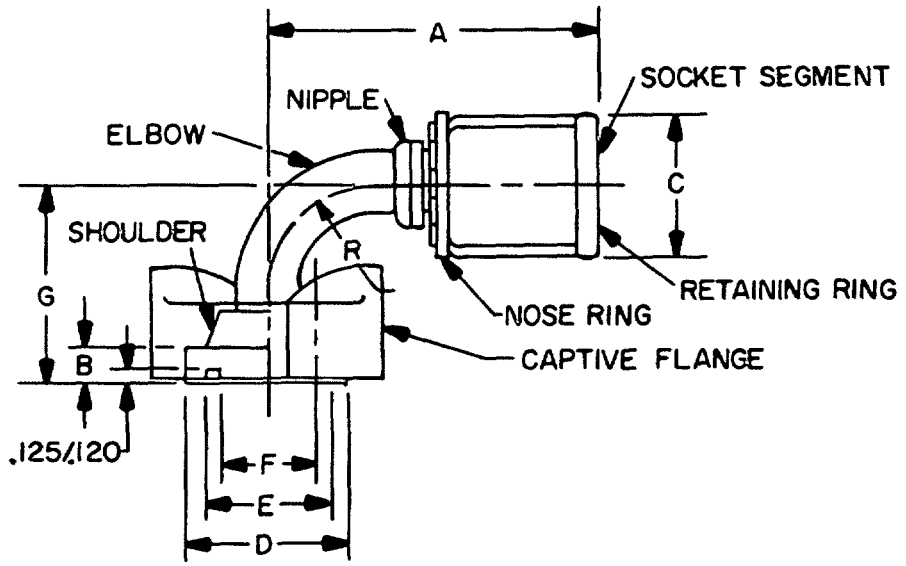
Size		Shoulder size	A	B	C	D	E	F	G	R
Od tube	Hose									
1-1/4	20	-20	6.68	0.405	2.88	2.125	1.752	1.422	1.96	1.75
1-1/2	24	-24	7.14	.495	3.16	2.500	2.120	1.890	2.06	2.00
2	32	-32	9.33	.495	3.75	3.125	2.495	2.165	3.06	3.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.
7. Captive flange material shall be malleable iron in accordance with ASTM A 220, grade 50005.

FIGURE 9. Type SAE J518c, flange fitting, 67-1/2 degree, with code 62 solid captive flange.

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Size		Shoulder size	A	B	C	D	E	F	G	R
Od tube	Hose									
1-1/2	20	-20	6.44	0.405	2.88	2.125	1.752	1.422	2.68	1.75
1-1/2	24	-24	7.25	.495	3.16	2.500	2.120	1.890	3.19	2.00
2	32	-32	8.95	.495	3.75	3.125	2.495	2.165	4.56	3.00

NOTES:

1. Dimensions are in inches.
2. Fitting assembly materials for the hose nipple shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished.
3. Socket for fitting assemblies for -20 hose and larger may be the one-piece or multiple segment design, not exceeding the envelope dimensions.
4. Fitting assembly materials for the socket shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished, or QQ-N-288, composition B.
5. Alternative material for the fitting assembly socket for hose sizes -20 and larger shall be steel conforming to SAE 4140 in accordance with AMS 6381, as applicable.
6. Tubing shall conform to MIL-T-16420, composition 70-30, grade 2, type I, class 1650 special.
7. Captive flange material shall be malleable iron in accordance with ASTM A 220, grade 50005.

FIGURE 10. Type SAE J518c flange fitting, 90-degree, with code 62 solid captive flange.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24135/9(SH)	Hose, Synethic Rubber, Wire Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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